Survey Weighting in the Cooperative Election Study

The Cooperative Election Study, the weights it applies to itself, and what a room can measure

Democracy's Data

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> *"Forty-two percent of Americans say…"*

Somebody asked some people, and the answer became a fact about a country of hundreds of millions. The step that turns the first thing into the second is **weighting**: counting some answers more than others, so that a sample which does not look like the country behaves as though it did. Almost every survey performs that step out of sight. This chapter is about the one large survey that performs it in the open.

The **Cooperative Election Study** interviewed about 60,000 people in 2024, which makes it one of the largest political surveys in the world. It publishes what almost no survey publishes: not only the answers, but the adjustments applied to them afterwards, side by side. Beside it, for scale, sit 19 people who answered seven of the same questions, in the same words, with the same answer options, in a single room on the first day of a term.

> **A caveat about the class column, before any of it is read.** The class file currently committed to this folder is `class_responses_EXAMPLE.csv`. **It is placeholder data. The responses in it are invented and are not yours.** They exist so that the tables have the right shape while the real first-day survey is being processed. Everything drawn from the CES is real and can be checked; nothing in the class column is, until the real file replaces the placeholder. Where this document reports a class figure, treat it as an illustration of the arithmetic and not as a fact about this room.

That caveat costs less than it looks like it costs, because the room is not the object of study here. The comparison is — and the most important thing the comparison turns up happens entirely inside the CES, on data that is real and checkable.

01 Where the data comes from, and what it is for

Nobody is required to run the Cooperative Election Study. It exists because dozens of university research teams each buy a module of it and share a common core of questions, coordinated since 2006 by Stephen Ansolabehere, Brian Schaffner and their collaborators. The release read here is the 2024 Common Content, Data Release 2, published on Harvard Dataverse with a DOI attached.

Who is in it, and how they got there. About sixty thousand American adults — and, crucially, not a random sample of them. The CES is an **opt-in online panel** of people who signed up with YouGov to take surveys. The panel is matched afterwards to a target sample drawn from a probability frame, so that it resembles the country, and then weighted. Sixty thousand respondents sounds like rigor, but size is not design, and everything this chapter shows follows from who joins a political survey panel voluntarily.

Who it was made for. Political scientists who need subgroups. A national poll of a thousand cannot say anything about one state or one small group; sixty thousand can, which is the whole purchase. The common content asks what political scientists have agreed to ask, in wording they have agreed on, and it would stop the year the money did.

What it costs to get. Nothing in money. The file is a free download from Harvard Dataverse for anyone with an account: about 60,000 rows and 694 variables, 184 MB. Those figures are typed rather than computed, because the respondent file is too large to ship here and is not this book's to redistribute. What ships here instead is a 39-row summary derived from it — one row per answer category of 7 questions, each carrying a headcount and the same share computed twice. The class file is not committed either, by design: it is one room's real answers, replaced in public copies by a placeholder with the same shape.

What has already been done to it. Almost everything. The CES teams drew the target sample, matched the panel onto it, wrote the wordings, assigned the numeric codes, validated voter registration against state records, and computed the weight, `commonweight`. The file arrives usable because a team spent years making it so, and their judgments arrive with it, unlabeled and inherited whole. The weighted column is not this chapter's arithmetic; it is somebody else's model, shipped as a column of numbers that looks exactly like the column of counts beside it.

02 How the data is published

A release appears on Dataverse in the months after each November election, and is revised once or twice as validation completes. Alongside it sit the questionnaires and the official Guide, the codebook that says what every numeric code means. Every percentage on this page began life as a column of single digits. Here is the class file, exactly as it is stored — the first line is its header, and the eight after it are eight students answering the same question:

"variable"	"code"
"gender4"	2
"gender4"	1
"gender4"	2
"gender4"	2
"gender4"	2
"gender4"	1
"gender4"	2
"gender4"	2

19 students, seven questions, 133 rows of that. There is no word anywhere in the file. `2` is not a quantity, and it does not mean the same thing in the `gender4` rows as in the `educ` rows below them. **The file is unreadable without a document that is not in the file**, and this is what almost every survey looks like when it arrives. The CES respondent file has the same shape at a different scale: one row per person, roughly 694 columns of digits, of which these are the seven this chapter reads.

Column	Codes that occur	What it asks
gender4	1..4	gender
educ	1..6	education
race	1..8	race
pid7	1..8	party ID (7 point)
ideo5	1..6	ideology (5 point)
newsint	1..4	news attention
votereg	1..3	voter registration

The Guide that decodes those digits has to be read carefully, because it once nearly wrecked this chapter. The Guide *displays* the race categories in a different order than their numeric codes run: it lists Middle Eastern sixth, while the category that actually carries code 6 is "two or more races". A build script that read the labels off the printed order would have produced this:

Code	By the Guide's printed order	By matching counts	Respondents
6	Middle Eastern	Two or more races	1,947
8	Two or more races	Middle Eastern	166

1,947 people called Middle Eastern and 166 called two or more races — a 12-fold error inside one category, in a table that would have looked entirely normal. Codes are therefore mapped to labels by matching the published category counts instead, **because the count is the only part of that mapping that can be checked against anything.**

Two files sit under this chapter, and they store different things.

File	One row is	Columns	Rows
ces2024_benchmarks.csv	one answer category of one question	variable, code, category, n, pct_unweighted, pct_weighted	39
class_responses(...).csv	one student's answer to one question	variable, code	133

The benchmark file is already a summary: not 60,000 interviews, but what those interviews produced, with the individual respondents gone. You can read the margins of any question and you can no longer cross one question against another. A summary is a decision about which questions can still be asked later, made before anyone knows what they will want. The class file is individual records, which sounds better and is not, because there are 19 of them.

03 What the weights do to a published number

A **weight** is a number attached to each respondent saying how many people like them the sample ought to have contained. If the panel caught half as many people without a high-school diploma as the country has, each one is counted roughly twice. The targets the weights aim at are counts of adults by age, sex, race and education. They come from the census and from federal surveys, which is how census data ends up quietly inside every survey estimate.

The rare thing about the CES is that both columns survive into print. Here is one row of the benchmark file:

Variable	Code	Category	Respondents	% unweighted	% weighted
votereg	1	Yes	54913	91.5	72.5

Read that row slowly. Two percentages, one set of 54,913 people. Nobody was asked twice and nobody was added. `% unweighted` is arithmetic on the interviews; `% weighted` is the sum of `commonweight` over the same people. The two numbers sit in adjacent columns, formatted identically, and one of them is a model.

Every CES figure anybody has ever seen has been through this, and the unweighted column is the only reason you can see how far it travelled. Here are the eight categories the weights move furthest:

Variable	Category	Unweighted (%)	Weighted (%)	Shift (points)
votereg	Yes	91.5	72.5	-19.0
votereg	No	7.4	23.2	+15.8
pid7	Strong Democrat	27.6	21.9	-5.7
newsint	Most of the time	51.2	46.4	-4.8
educ	No HS	3.6	7.5	+3.9
votereg	Don't know	1.1	4.3	+3.2
ideo5	Not sure	6.0	9.2	+3.2
educ	Some college	23.3	20.4	-2.9

The largest adjustment in the file is voter registration. 91.5% of CES respondents said they were registered to vote. The weights cut that to 72.5% — **19.0 points.** Nobody lied. The adjustment is not correcting an answer; it is correcting a sample. People who join an online political panel are far more likely to be registered than people who do not, and the weight is a formal confession of that fact.

The same mechanism runs down the table. Strong Democrats fall 5.7 points, and people who follow the news "most of the time" fall 4.8. People with no high-school diploma, the group least likely to opt into a political panel, rise 3.9. Summed question by question, the weights move 19.0% of the sample on voter registration, more than on any political question. The panel's problem is engagement, not opinion, and the weights say so.

Now watch that machinery move a headline. Seven-point party identification collapses naturally into three groups. Do it to both columns.

Group	CES unweighted (%)	CES weighted (%)
Democrat	49.1	41.6
Independent	13.2	15.3
Not sure	2.1	3.4
Republican	35.6	39.7

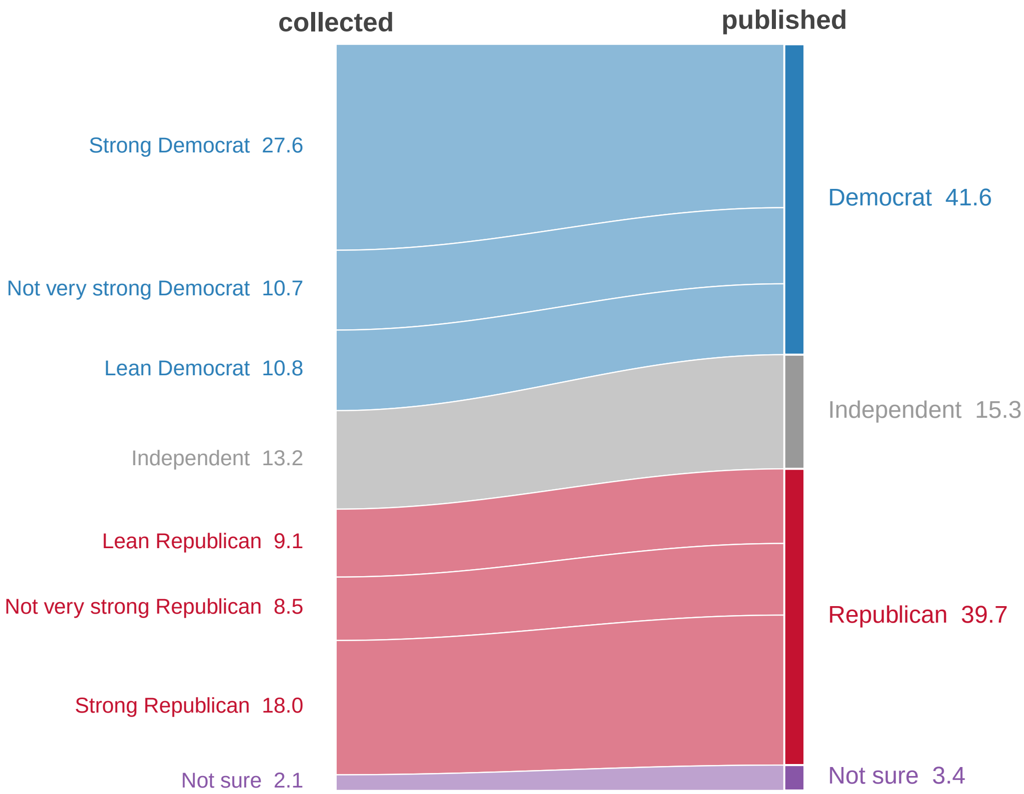

Figure 1. Seven answer options collapsing into three groups, and the weights changing the headline on the way. Each ribbon is one answer option, entering on the left at the width the researchers collected and leaving on the right at the width they publish; ribbon width is the share of the sample. Ribbons fit this figure because the two columns are the same people counted twice: nothing enters or leaves, it only thickens and thins. A ribbon is the one chart form that shows a reallocation without pretending anything moved in or out. The Democratic block enters at 49.1% and leaves at 41.6%; the Republican block enters at 35.6% and leaves at 39.7%. Hover a ribbon for the answer option behind it.

The sample the CES collected is D+13.5. The sample it publishes is D+1.9. The weights move the headline partisan balance of the largest political survey in America by 11.6 points. That is not a scandal — the weighted figure is very likely the better estimate of the country. It is a demonstration that a published survey number is a construction, built by people who had to decide which variables mattered, and that the deciding happened before you ever saw it.

04 A room is a census of itself

The class survey has no weights. Nobody built any, and nobody could: **a weight requires a population the sample is a sample of**, and there is no population of which this room is a sample. The room is a census of itself, so the class column is the equivalent of the unweighted CES column — raw, and describing whoever was there. Here it is against the country, on the question everybody wants:

Party identification	Class (%)	CES unweighted (%)	CES weighted (%)
Strong Democrat	26.3	27.6	21.9
Not very strong Democrat	10.5	10.7	10.2
Lean Democrat	21.1	10.8	9.5
Independent	31.6	13.2	15.3
Lean Republican	5.3	9.1	10.0
Not very strong Republican	0.0	8.5	9.6
Strong Republican	5.3	18.0	20.1
Not sure	0.0	2.1	3.4

Reminder: the first column is placeholder data, not this class. The two CES columns are real.

The instinct is to read across the rows and start explaining the differences, and the instinct is wrong for a second reason that has nothing to do with weights.

Survey	Respondents	Margin of error (± points)
This class	19	22.5
A typical national poll	1,000	3.1
CES 2024	60,000	0.40

With 19 respondents, the margin of error on any percentage the class produces is about **±22.5 points** — how far off the estimate could be, and still be doing its job. A fifteen-point class-versus-country gap is not a finding; it is inside the noise. Across all 39 class-versus-CES comparisons these seven questions support, only 2 clear that band, and the other 37 sit where the room and the country cannot be told apart. Getting the room to a respectable ±3 points would take about 56 times as many people, because precision improves only with the square root of the sample.

So the two halves of this chapter fail in two different ways, and the reflex to conflate them is the thing worth unlearning.

Problem	What goes wrong	Where you saw it	Does a bigger sample fix it	Does weighting fix it
Bias	the sample is systematically unlike the population	CES: registration off by 19.0 points before weighting	No	Only for variables you measured

Problem	What goes wrong	Where you saw it	Does a bigger sample fix it	Does weighting fix it
Imprecision	the sample is too small to resolve anything	this class: ±22.5 points on every estimate	Yes, slowly	No

Neither one fixes the other. A bigger sample makes you more precise about whatever your sample happens to be. Weighting corrects the imbalances somebody thought to measure, and nothing else. The 19.0-point registration adjustment and the ±22.5-point margin are two entirely different kinds of wrongness, and only one of them gets mentioned when a survey reaches a newspaper.

None of that is yet a statement about any particular room, because the class file in use is the placeholder. The arithmetic is correct; the responses it was applied to are invented.

05 What this data cannot tell you

Whether the weighted column is right. A weight can only correct an imbalance on something that exists in both the panel and the census tables it aims at. Whatever made someone join an opt-in panel is not a column in any census table, so it can be corrected only as far as it travels with the demographics that are. Which variables the weights are still missing is, by construction, not in the file, and never will be. The AAPOR review of the 2016 polls found exactly this failure at scale: state polls over-represented college graduates, mostly did not weight on education, and that single unadjusted variable explains much of the miss.

Whether opt-in panels beat probability samples. Ansolabehere and Rivers make the case that a large, matched, transparently weighted panel now beats a probability sample that almost nobody answers; the AAPOR reviews supply the caution. Groves adds that a response rate on its own settles nothing, because nonresponse bias depends on whether answering correlates with the thing measured — which is precisely the shape of the CES's own registration adjustment. Serious people hold both positions, and everything above is compatible with either.

Anything that needs the individual respondents. The benchmark file here is a summary, so there is no cross-tabulation: you cannot ask whether a gap survives holding age constant. Item non-response is present and invisible throughout — 1,478 people skipped the news-attention question, and no published figure mentions them.

Anything confident about the room. 19 respondents cannot resolve a difference smaller than about 22.5 points, and almost every interesting difference is smaller than that. Mode differs too: the CES is taken alone at home, the class survey in a room where an instructor could see the screen — identical words, a different instrument. And nobody can say what the room would look like weighted, because there is nothing to weight it to.

To which one more must be added here: **the class responses in use are placeholder data**, invented to give the tables their shape. Every class figure in this document is illustrative only.

06 What you should have learned

- A weight says how many people like this respondent the sample should have contained. It is the statistical method that makes a sample representative, and census figures are the targets it aims at.
- The CES publishes both columns, so the adjustment is visible: registration moves 19.0 points, and the collapsed partisan margin moves from D+13.5 collected to D+1.9 published.
- The largest adjustments all describe engagement with politics rather than opinion about it, which is what self-selection into a political panel predicts.
- Weighting fixes only what somebody measured; a bigger sample fixes only imprecision. Bias and noise are different failures, and neither remedy touches the other.
- A complete count of a room has no margin of error and no weights, because there is no population it samples — and at 19 people it resolves nothing finer than about ±22.5 points anyway.

07 Where this data appears in this book

- Sixty Thousand People, and Sixty-Eight Vermonters — what the CES's size buys state by state, and where even sixty thousand runs out.

The section's opening chapter, what a survey can establish, reprints this chapter's weighting benchmarks as its evidence that the repair is large.

08 Sources

Data

- Cooperative Election Study, 2024 Common Content, Data Release 2 (Ansolabehere, Schaffner and Pope), Harvard Dataverse. https://doi.org/10.7910/DVN/X11EP6. The benchmark file is derived from it; the weight is `commonweight`, and the Guide is the codebook every numeric label above was checked against.
- Class replication survey, first day of term, recoded to CES codes. Not redistributed; the committed placeholder has the same shape.

The data itself

Every figure above is drawn from these tables. They are plain CSV, they open in a spreadsheet, and they are yours to take further.

`ces2024_benchmarks.csv`, `class_responses_EXAMPLE.csv`

Your turn

None of these is answered above, and every one can be answered from the tables you just opened.

- `ces2024_benchmarks.csv` reports `pct_unweighted` next to `pct_weighted` for every category. Sort by the difference. Which groups is the weighting working hardest to fix?

- Take one variable in `ces2024_benchmarks.csv` and find the category the raw sample is furthest off on. What kind of person is hardest to get to answer a survey?
- `class_responses_EXAMPLE.csv` is one room's answers in the same coding. Compare a variable in it against the national benchmark. Where is a classroom least like the country?
- Weighting fixes what it can see. Name something about a respondent that no weight in this file could correct for.

Academic literature

- Ansolabehere, S. and Rivers, D. (2013). "Cooperative Survey Research." *Annual Review of Political Science* 16: 307–329.
- Groves, R. M. (2006). "Nonresponse Rates and Nonresponse Bias in Household Surveys." *Public Opinion Quarterly* 70(5): 646–675.
- Kennedy, C. et al. (2018). "An Evaluation of the 2016 Election Polls in the United States." *Public Opinion Quarterly* 82(1): 1–33.

WHY THIS DATA CANNOT BE FETCHED AGAIN

The file behind this chapter comes from a source no script can fetch today, though a person with a browser can.

WHY NOT. The chapter is built on the 2024 Cooperative Election Study Common Content — 60,000 Americans interviewed before and after the 2024 election — held on Harvard Dataverse at

https://doi.org/10.7910/DVN/X11EP6

Dataverse sits behind an AWS web application firewall that answers an automated request with status 202, "Accepted", and a small challenge page instead of the data. The code says success; the body is a wall only a real browser can pass. And at 184 MB the survey file is too large to travel with this book in any case.

WHAT EXISTS INSTEAD. Open the address above in an ordinary browser and the download works normally; the data are free. The benchmark table this chapter actually uses — every category of seven core questions, with its unweighted share and its share under the study's own weight — is small and ships alongside, so nothing needs downloading to follow the argument.

WHAT YOU CAN STILL CHECK. If you do download the survey file, these numbers say whether you hold the same object:

- 60,000 rows and 694 columns
- 54,913 respondents said they were registered to vote — 91.5% of

> the sample unweighted, and 72.5% once the weight is applied. That nineteen-point correction is the chapter’s point.
>
> – 16,576 called themselves strong Democrats and 10,826 strong Republicans, before any weighting